

EXPERIMENT 17 - WRITE A PROLOG PROGRAM TO SUM THE INTEGERS FROM 1 TO N

Aim

To develop a Prolog program to calculate the sum of all integers from 1 to a given positive integer N using recursion.

Objective

- To understand the basic concepts of Prolog programming.
- To implement recursion in Prolog.
- To calculate the sum of integers from 1 to N.
- To understand how facts, rules, and queries are used in Prolog.

Theory

Prolog is a logic programming language commonly used in Artificial Intelligence. Instead of specifying a sequence of instructions, a Prolog program defines facts and rules, and the Prolog interpreter determines the solution based on logical inference.

The sum of integers from 1 to N can be calculated using the recursive formula:

$$\text{Sum}(N) = N + \text{Sum}(N-1)$$

The base condition is:

$$\text{Sum}(0) = 0$$

For example:

$$\text{Sum}(5) = 5 + 4 + 3 + 2 + 1 = 15$$

The program uses recursion to repeatedly reduce N until it reaches 0.

Algorithm

Step 1: Start the program.

Step 2: Define the base case as `sum(0, 0)`.

Step 3: Define the recursive rule to add N to the sum of integers from 1 to N-1.

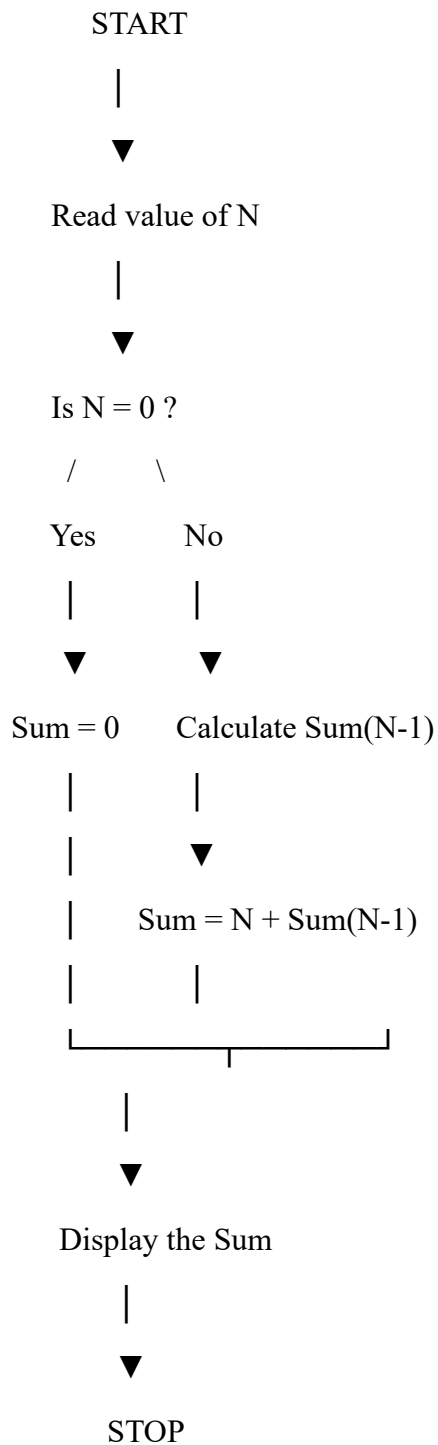
Step 4: Accept a value of N through a Prolog query.

Step 5: Recursively calculate the sum.

Step 6: Display the calculated sum.

Step 7: Stop the program.

Flowchart



Prolog Program

% Base case

sum(0, 0).

% Recursive case

sum(N, S) :-

$N > 0$,
 $N1$ is $N - 1$,
 $\text{sum}(N1, S1)$,
 S is $N + S1$.

Query

?- $\text{sum}(10, S)$.

Sample Output

$S = 55$.

Another example:

?- $\text{sum}(5, S)$.

Output:

$S = 15$.

Explanation

The predicate $\text{sum}(N, S)$ calculates the sum of integers from 1 to N .

- $\text{sum}(0, 0)$ is the base case. When N becomes 0, the sum is 0.
- $N1$ is $N - 1$ decreases the value of N by 1.
- $\text{sum}(N1, S1)$ recursively calculates the sum up to $N-1$.
- S is $N + S1$ adds the current value of N to the previously calculated sum.

For example, for $N = 5$:

$\text{sum}(5, S)$
 $= 5 + \text{sum}(4)$
 $= 5 + 4 + \text{sum}(3)$
 $= 5 + 4 + 3 + \text{sum}(2)$
 $= 5 + 4 + 3 + 2 + \text{sum}(1)$
 $= 5 + 4 + 3 + 2 + 1 + \text{sum}(0)$
 $= 15$

Result

The Prolog program was successfully implemented to calculate the sum of integers from 1 to N using recursion. For the query $\text{sum}(10, S)$, the program successfully produced the result $S = 55$.

Conclusion

Thus, the Prolog program successfully calculates the sum of integers from 1 to N using recursive rules. This experiment demonstrates the use of recursion, arithmetic operations, predicates, and logical inference in Prolog programming.

```
3
4  def sum_numbers(n):
5      # Base case
6      if n == 0:
7          return 0
8
9      # Recursive case
10     return n + sum_numbers(n - 1)
11
12
13     # Get input from the user
14     n = int(input("Enter the value of N: "))
15
16     # Calculate and display the sum
17     result = sum_numbers(n)
18
19     print("Sum =", result)
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Varshini M\AppData\Local\Programs\Microsoft VS Code> & "C:\Use
ni M/OneDrive/Desktop/Here!!!/ARTIFICIAL INTELLIGENCE/VSC EXPs/EXP1.py"
Enter the value of N: 5
Sum = 15
PS C:\Users\Varshini M\AppData\Local\Programs\Microsoft VS Code> |
```